Three recordings, three acoustic profiles

Measured 30-second excerpts · separate stereo channels · no clinical inference

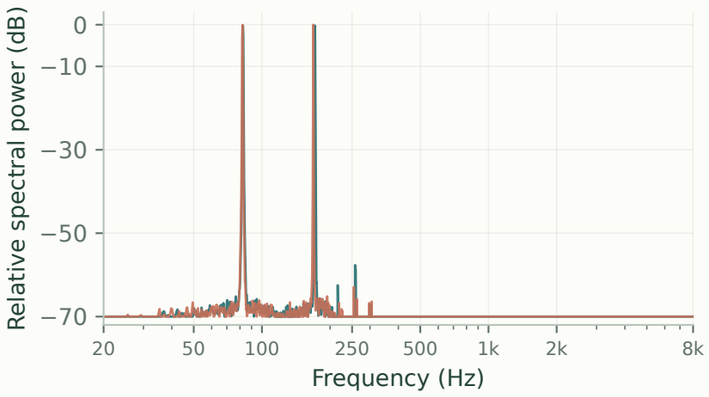
Left channel Right channel

Frequency content

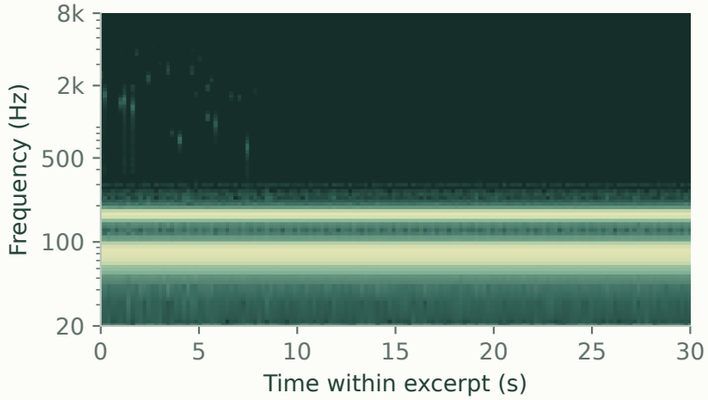
Time–frequency structure

Level over time

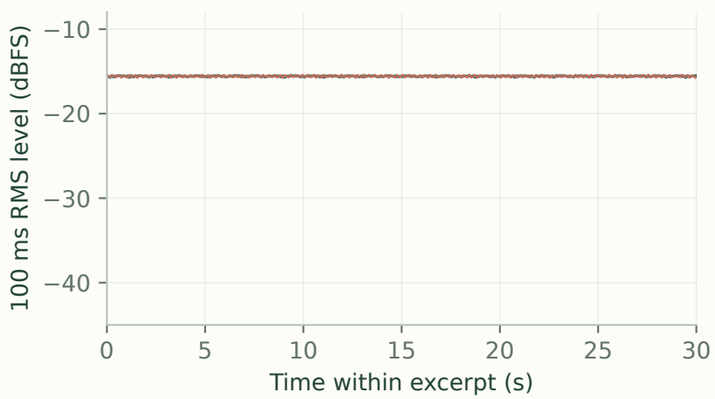
Binaural reference · 32:48–33:18



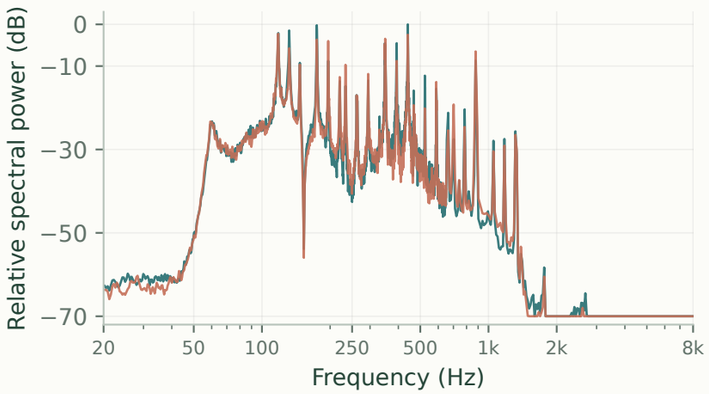
L/R power averaged



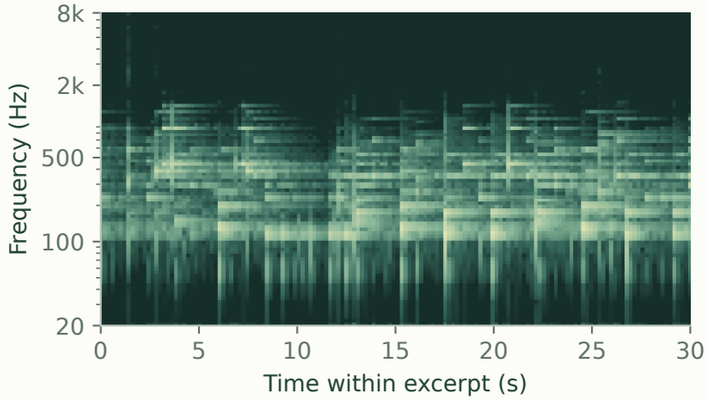
Centroid 126 Hz · low-band energy 100%



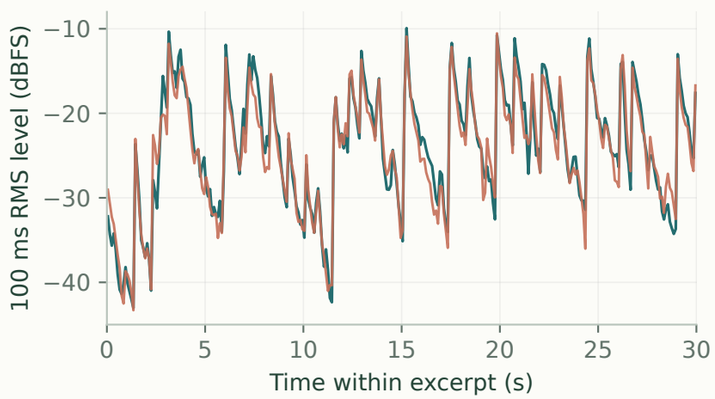
Handpan reference · 01:00–01:30



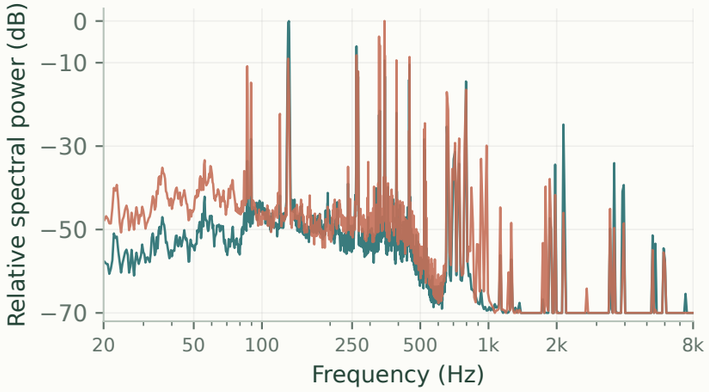
L/R power averaged



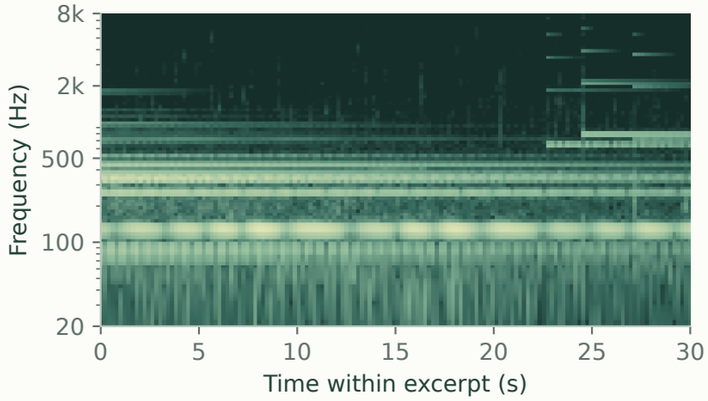
Centroid 303 Hz · low-band energy 51%



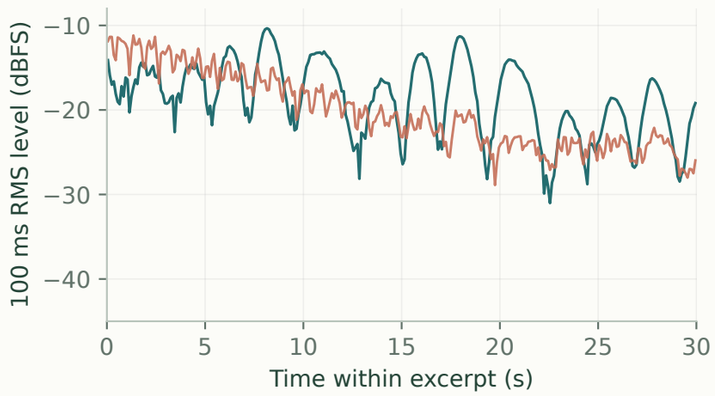
Singing-bowl reference · 01:00–01:30



L/R power averaged



Centroid 236 Hz · low-band energy 54%





Methods: spectra average 131,072-sample Hann windows (50% overlap); 150 sampled short windows form each spectrogram.

Spectra and spectrograms each use an independent 0 dB reference per recording. Colors do not compare absolute loudness across recordings.

dBFS is a digital level, not ear-level dB SPL. These plots describe sound—not brainwaves, migraine relief, or a personal “sweet spot.”